Measuring Orthographic Retrieval Failure in Children: A Cross-Validated 30-Item Test of Chinese Character Amnesia

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Abstract

Background. Written language production requires rapid access to orthographic representations and their mapping to motor plans. In Chinese, this mapping is especially demanding because character production depends on retrieving visually complex forms composed of radicals and stroke sequences (Tong & McBride-Chang, 2010). A growing number of Chinese school-age children nowadays suffer from character amnesia: difficulty in writing a character correctly, despite being able to recognize it. On the other hand, there is currently no standardized diagnostic tool for character amnesia targeting school-age children. We tackled this issue by conducting a large-scale survey on character amnesia among primary school students in China and subsequently developed an assessment measure of orthographic retrieval failure for school-age children.

Method. Forty Grade-5 children (23 female; mean age of 10) completed a handwriting-to-dictation task in simplified Chinese. Eight hundred target characters were selected from an adult handwriting database (Xu et al., 2025) using frequency (log 1.5-5.0), stroke-count (>4), child curriculum (pre-Grade 5) and context-word familiarity constraints, yielding a final set of 800 characters (mean log frequency 3.55; 5-20 strokes; 1-7 radicals). Using OpenHandWrite with a Wacom tablet, each child wrote 200 characters according to the audio dictation-prompt across four sessions. After each trial of dictation (Figure 1), children self-reported "correct," "character amnesia," or "don't know." Two helpers verified all pen scripts; three participants with high questionable self-reports were excluded. Analyses used only correct vs. amnesia responses.

The diagnostic measure of Chinese character amnesia was built upon the handwriting-to-dictation task by selecting a subset of Chinese characters of outstanding psychometric qualities. First of all, characters with near-zero variance in children's response were removed (amnesia rate < .01; 360 items), leaving 440 candidates. Then we built a IRT model in Stan (Stan Development Team, 2024) which treated each trial as a Bernoulli random variable with probability of success given by Pr(success) = $\text{logit}^{-1}(\gamma j(\alpha_i-\beta_j))$ for participant i responding to item (character) j, where the participant has ability α, and the item has difficulty β, and discrimination γ. Four item (character) selection methods were compared: random baseline, maximum γ, diverse difficulty (β-binned with top-γ per bin), and an upper-lower thirds discrimination score, which rank items by how well they separate the highest and lowest 1/3 of participants. Subset sizes (4-70) were evaluated by correlation with full-test amnesia rates (Figure 2). Crucially, we evaluate candidate item subsets by out-of-sample prediction: In each of 30 repetitions, we sampled 10 participants and held out their data. Item characteristics were estimated using the data of the remaining 25 participants, and we selected item subsets according to the above four selection procedures. We then compared the amnesia rates of the 10 held-out participants estimated from the 30-item subsets with their observed ground truth amnesia rates based on the full set of 800 items (Figure 4).

Results. Among multiple selection schemes, the upper-lower thirds discrimination procedure yields a compact 30-item test (Figure 3) that preserves individual-difference structure and generalizes to unseen test-takers (cross-validated mean r =.74 with full-test rates; within-sample r =.93 at 30 items).

Discussion. This work contributes to psycholinguistics by providing a process-sensitive measure of children's ability to retrieve orthographic forms during production (a component minimized in copying tasks) and encourages further research on how lexical frequency and orthographic complexity constrain written production. Second, it introduces a validation process that aligns psychometric modeling with language processing research. The resulting short test provides a scalable tool for identifying profiles of children consistent with retrieval-based writing difficulties.

References

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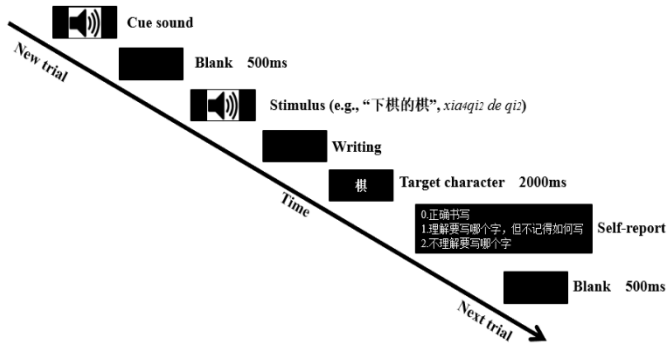


Figure 1. The experimental schema was adapted from Xu et al. (2025). It began with a cue sound "Ding", followed by a 500ms blank interval. Participants then heard the dictation prompt and could immediately begin writing. This was followed by a 2000ms presentation of the character that was to be written. Participants provided self-report responses by pressing the numbers 0 (correct handwriting), 1 (character amnesia), or 2 (don't know) on the keyboard. Another 500ms blank interval followed before the next trial.

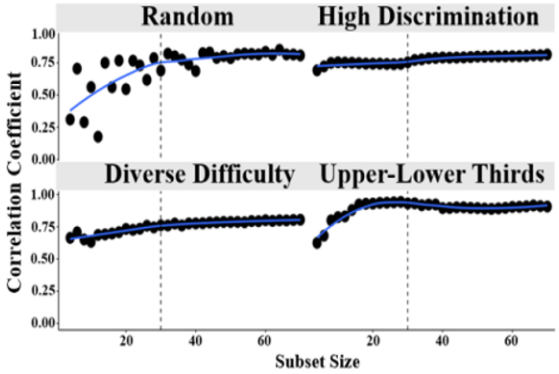
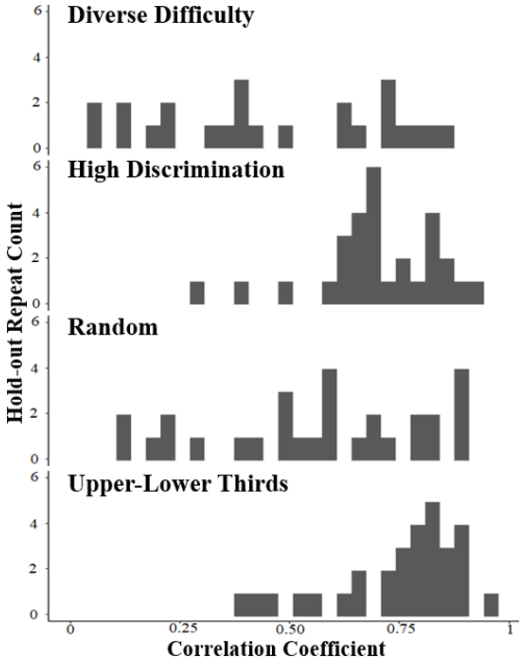


Figure 2. Correlations between predicted and observed character amnesia rates using test subsets selected according to the random, maximum discrimination, diverse difficulty and the and upper-and-lower-thirds method. A proposed test-set size of 30 items is highlighted with a dashed vertical line.



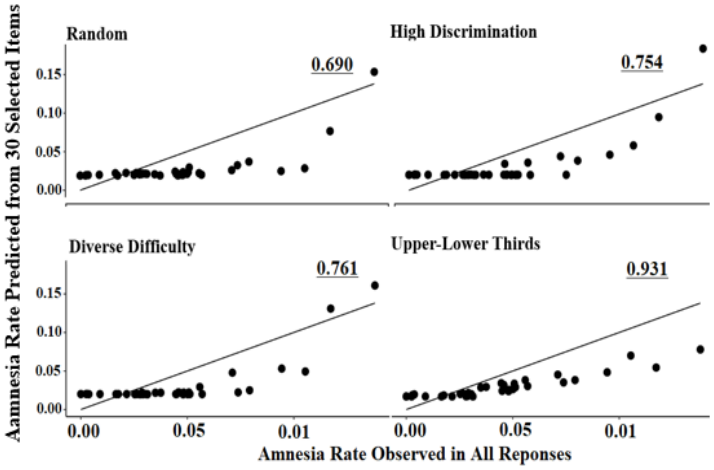


Figure 3. A single prediction check at the test size of 30 items

Figure 4. Histograms show the 30 correlations obtained between estimates of the 10 hold-out participants' amnesia rates based only on the selected subset of items and the observed amnesia rates taken from their responses to all items.